

1. Compile the code

It is very easy to compile the code. In the Linux environment, just typing 'gfortran -o gmls @compiler.txt' in the command line can get the desired executive file 'gmls'. Since it is a very simple code, there is no make file. In addition, if you would like to use compilers other than gfortran, you can also type the corresponding command line. The name of this executive file (gmls here) can also be changed.

It is noted that sometimes you may see errors reported by the system when you first compile this set of code, because the modules 'define.mod', 'point.mod', 'kd_tree.mod', have not been compiled. You can just retype in the command line again to get the executive file (errors are reported in the first compiling, but the mod files are generated, so compiling the second time will get the right file).

2. Function of each subroutine

This set of code consists of four files, where 'k_d_tree.f95' is the k-d tree library written by Matt Kennel, so I do not know the details but only incorporate that into the code. The other parts are written by me.

The file 'main_io.f95' includes the main program and the I/O files; the file 'adv.f95' is the advection part; the file 'GMLS_diff.f95' is the part using the GMLS to solve the dispersion process. Some explanations have been added into the code, but please feel free to let me know if there are any questions.

If learning the GMLS method is the focus, I would suggest reading the subroutine 'MATRIX_INV_POLY' in the 'GMLS_diff.f95' first, which is specifically written for the GMLS method. The LU decomposition method, written as the subroutine 'LU_INVERSE', is adopted to compute the inverse of the matrix. I have a version using the SVD method to complete this calculation, but it is somehow much slower than the LU decomposition due to the computation of the singular values. Since the P-unisolvence is typically required in the GMLS method, I employ the LU decomposition here but have a part to warn the singularity.

3. Function of each input file

In the folder 'test_file', the input files of an example can be found. The function of each file is: (a) 'basic_input.dat' is the input file for the basic parameters; (b) 'head_ini.dat' is the input file for the initial concentration (this name is kind of legacy inherited from my previous code written for the groundwater hydrodynamics); (c) 'velocity.dat' is the input file for the velocity field;

(d) 'velocity_ini.dat' inputs the parameters for the interpolation of the velocity (from the input field to each particle); (e) 'xyz.geo' is the coordinates of each particle; (f) 'output_time.dat' determines which time step to output the results.

4. Format of input files

Some explanations have been included in the code to demonstrate the meaning of each variable, so here only the corresponding part in the code for each input file is shown.

a) For the 'basic_input.dat'

The corresponding code to read that is from lines 173-181 in 'main_io.f95', which is shown below:

```

169 CALL getcwd(PATH)
170 PATH = trim(PATH)
171 NAME1 = trim(PATH)
172 OPEN (1,FILE = trim(NAME1))
173 READ (1,*) XYZ_TNUM,XYZ_NUM ! Total Points, Total Local Points
174 READ (1,*) DT_INPUT_TYPE ! Time step type
175 READ (1,*) T_TYPE ! Whether it is uniform flow
176 READ (1,*) T_LIM,DT,DT_AMP,T_START ! Time Limits, Time Step, Amplification, Starting time
177 READ (1,*) NCYC_OUT,NCYC_CHECK
178 READ (1,*) HEAD_NUM,FLUX_NUM ! HEAD_NUM is the total number of the essential BC
179 READ (1,*) DRAW_NUM ! The total number of the output times
180 READ (1,*) ALPHA_T,ALPHA_L,DM ! Coefficients for the dispersion along the transvers and longitudinal directions. DM is the diffusion coefficient
181 READ (1,*) NAME_FILE ! Name of the case
182 CLOSE (1)
183

```

Some detailed explanations: when 'DT_INPUT_TYPE = 0', the time step will be the manually input one from 'DT'; otherwise, it will be the minimal value between DT and the criterion for the stable time step. 'T_TYPE' is determined by whether the flow is uniform (= 0 for uniform flow) since for the uniform flow it is not needed to recompute the k-d tree as well as the GMLS matrix.

b) For the 'head_ini.dat', the corresponding code is from lines 78-82 in 'main_io.f95'.

c) For the 'velocity.dat', the corresponding code is from lines 205-210 in 'main_io.f95'.

d) For the 'velocity_ini.dat', the corresponding code is from lines 198-202 in 'main_io.f95'.

e) For the 'xyz.geo', the corresponding code is from lines 222-227 in 'main_io.f95'.

f) For the 'output_time.dat', the corresponding code is from lines 214-219 in 'main_io.f95'.

5. Output files

The output part of the code is from lines 132 to 149 in 'main_io.f95' for the concentration (the name convention is in line 134) and dispersion coefficient at each particle (name convention is line 143). The CPU time is output in line 158.